MOTOR PATHWAY

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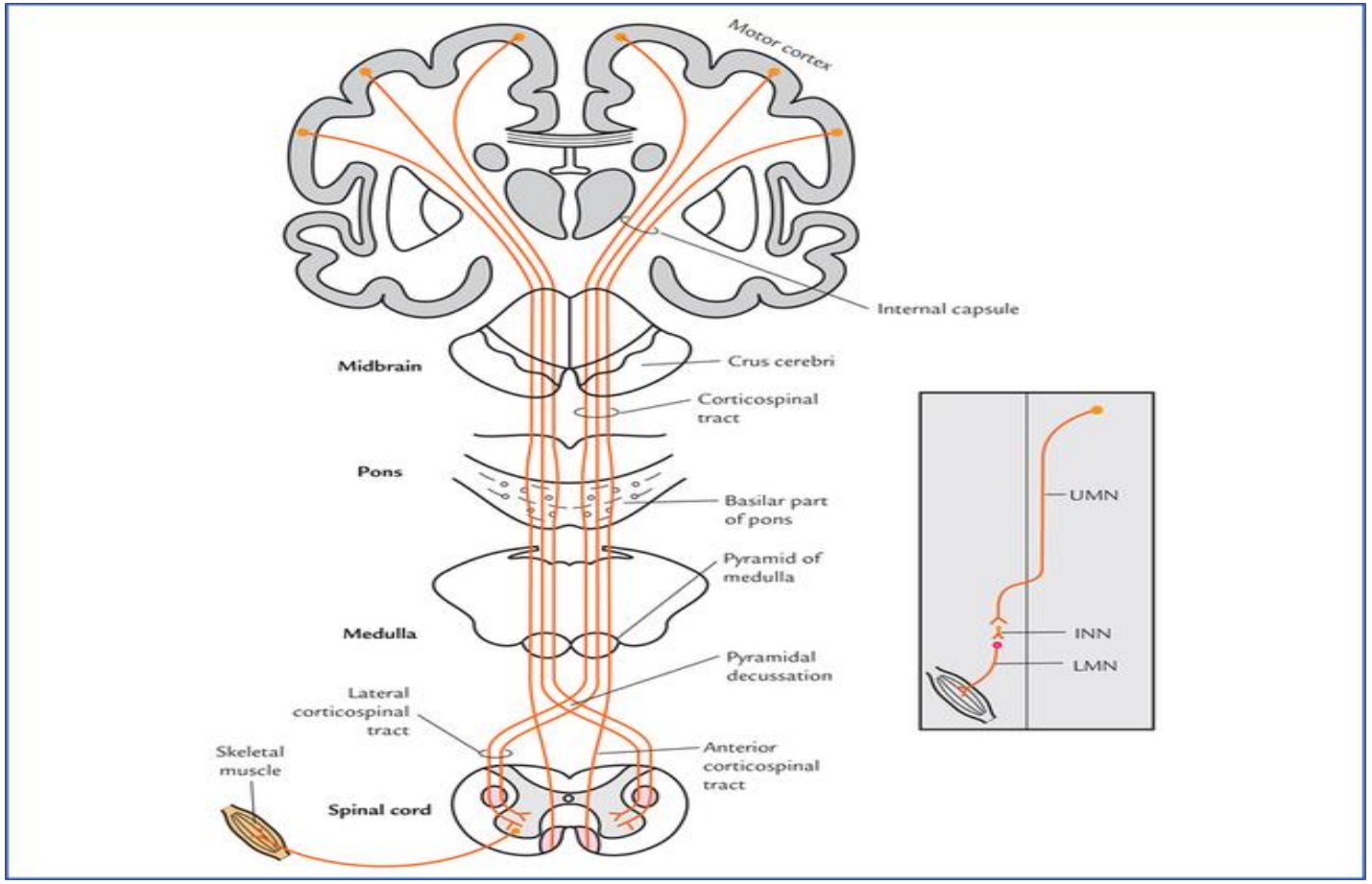
OUTLINE

- INTRODUCTION
- COMPONENTS
- PYRAMIDAL TRACTS
- EXTRAPYRAMIDAL TRACTS
- UPPER MOTOR NEURONS (UMN)
- LOWER MOTOR NEURONS (LMN)
- FUNCTIONS
- APPLIED ANATOMY
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INTRODUCTION

- The motor pathway are nerve pathways through which signals are transmitted from the brain to skeletal muscles
- They control both voluntary and involuntary movements
- The motor pathway involve two main tracts which are pyramidal and extrapyramidal tracts
- There are two major neurons involved in the transmission of this signals, they include; upper motor neurons (UMN) and lower motor neurons (LMN)

THE MOTOR PATHWAY



COMPONENTS

- The motor pathway is made up of the followings;
- Pyramidal tracts
- Extrapyramidal tracts
- Upper motor neurons (UMN)
- Lower motor neurons (LMN)

PYRAMIDAL TRACTS

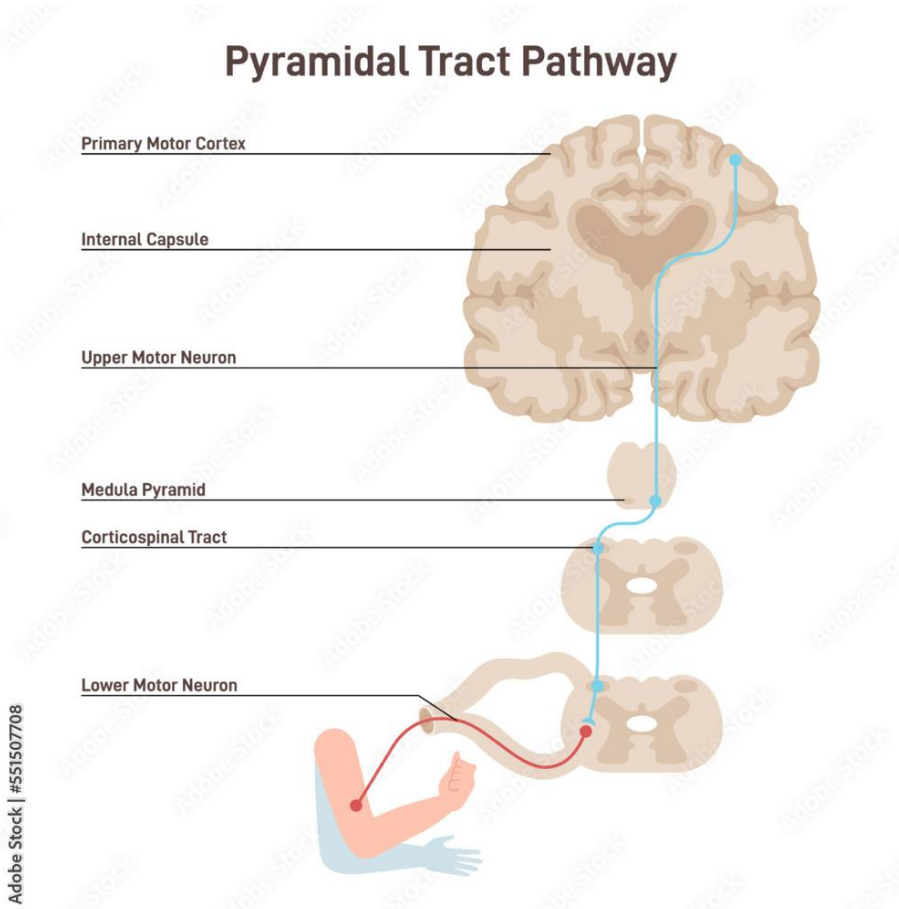
- Definition; the pyramidal pathway is the main voluntary motor pathway
- It include;
- i. Corticospinal tract- supplies the musculature of the body
- ii. Corticobulbar tract- supplies the muscles of the head and neck
- Origin; the fibers of the tract originates from a wide area of the cerebral cortex such as-
- (a). Primary motor areas give rise to about two-third of the fibres
- (b). Premotor area of the frontal lobe give rise to about one-third
- (c). Supplementary motor area

The somatosensory motor areas also give rise to some fibers that regulates the activities of the descending tracts

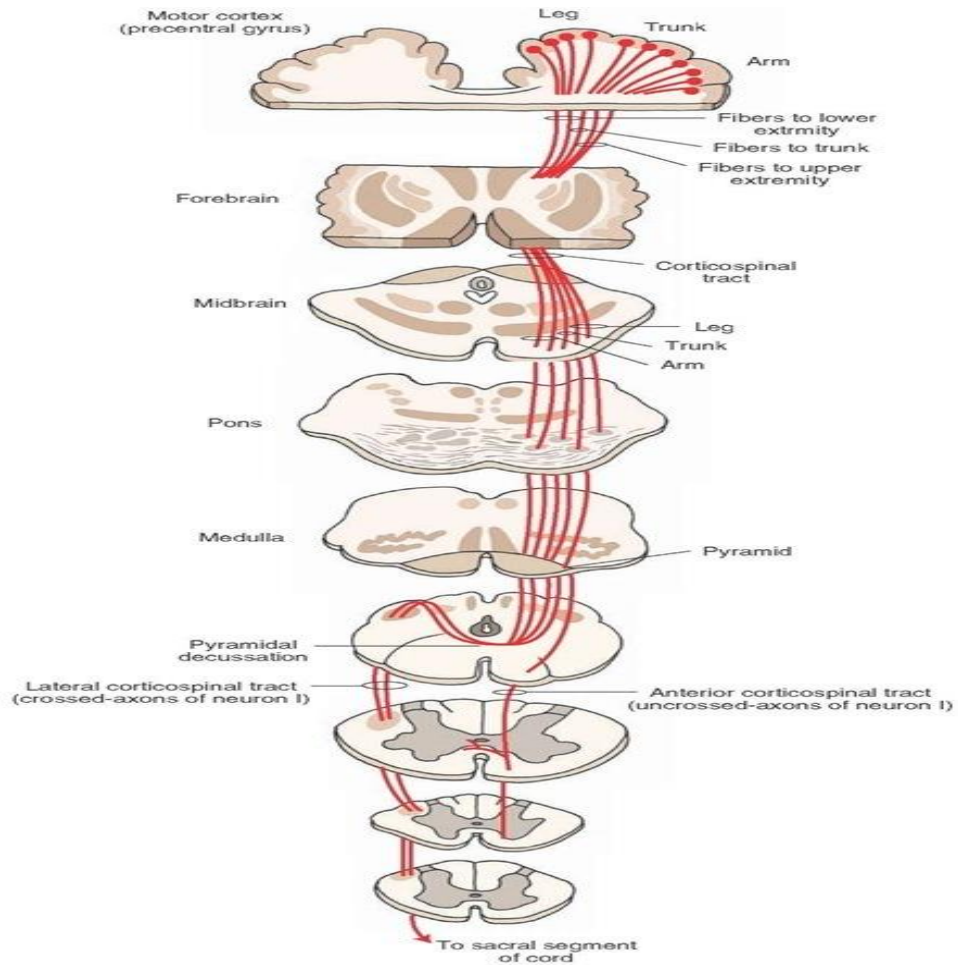
PYRAMIDAL TRACT Cont'd

- Course; From the cerebral cortex, it passes through the corona radiate, through the genu and posterior limb of the internal capsule
- The fibers then enter the crus cerebri of the midbrain where they occupy the middle two-thirds of the crus
- They then descend through the ventral part of the pons to enter the pyramid in the upper part of the medulla
- Near the lower end of the medulla, about 80% of the fibers cross to the opposite side and occupy a central position in the lateral white column of the spinal cord.
- The crossed fibers constitute the decussated (crossed) pyramidal tract, the lateral corticospinal tract
- A small portion remains uncrossed until they reach the level of the spinal cord where they terminate (about 20%), this is the uncrossed pyramidal tract, the anterior corticospinal tract
- Termination; the pyramidal tract ends in the grey matter at various levels of the spinal cord

DIAGRAM


Pyramidal Tract Pathway
Primary Motor Cortex
Internal Capsule
Upper Motor Neuron
Medula Pyramid
Corticospinal Tract
Lower Motor Neuron
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DIAGRAM


Motor cortex
(precentral gyrus)
Leg
Trunk
Arm
Fibers to lower extrmity
Fibers to trunk
Fibers to upper extremity
Forebrain
Corticospinal tract
Midbrain
Leg
Trunk
Arm
Pons
Medulla
Pyramid
Pyramidal decussation
Lateral corticospinal tract
(crossed-axons of neuron I)
Anterior corticospinal tract
(uncrossed-axons of neuron I)
To sacral segment of cord

CORTICOBULBAR TRACT

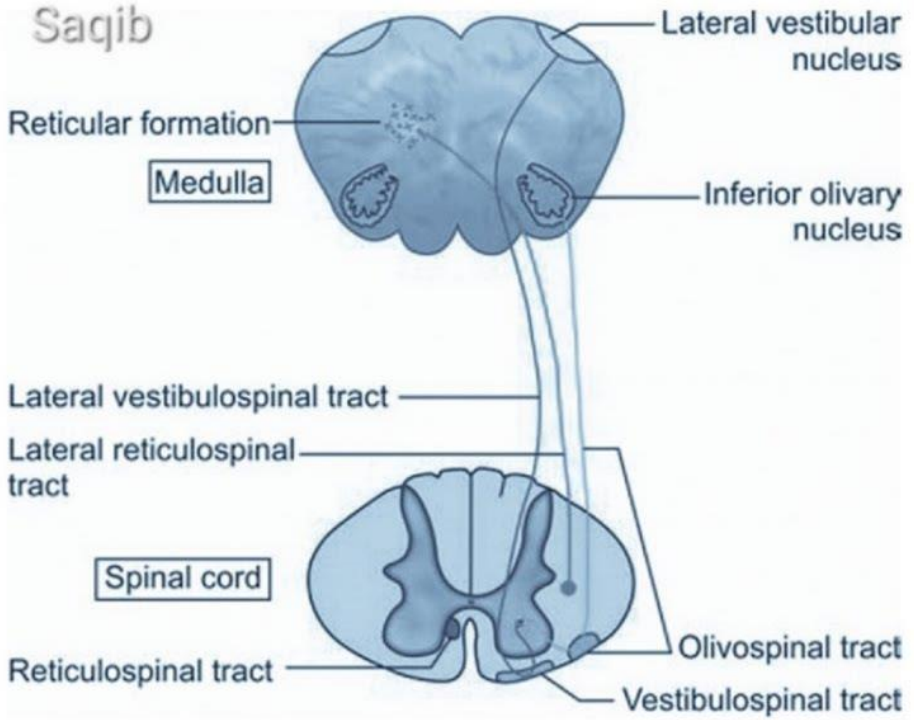
- Origin; It arise from the lateral side of the primary motor area of the cerebral cortex
- Receives the same inputs as the corticospinal tracts
- The fibers converge and pass through the internal capsule to the brainstem
- Termination; the fibers terminate on the motor nuclei of the cranial nerves, where they synapse with the lower motor neurons, which conveys the motor signal to the muscles of the head and neck
- Most fibers innervate the motor neurons bilaterally

EXTRAPYRAMIDAL TRACTS

- The extrapyramidal tracts are motor pathways that originate in the brainstem
- They carry motor fibers from the brainstems to the spinal cord
- Responsible for the involuntary and autonomic control of all musculature, such as muscle tone, balance, posture and locomotion
- There are four tracts that constitute the extrapyramidal tracts;
- i. Vestibulospinal tract
- ii. Reticulospinal tract
- iii. Rubrospinal tract
- iv. Tectospinal tract
- V. olivospinal tract
- The vestibulospinal and reticulospinal tract do not decussate, they provide ipsilateral innervations while the rubrospinal and tectospinal tracts decussates and provides contralateral innervations

DIAGRAM

EXTRAPYRAMIDAL TRACTS



VESTIBULOSPINAL TRACTS

- Origin; from the vestibular nuclei
- Receives input from the organs of balance
- Then transmits the impulse for balance to the spinal cord
- There are two vestibulospinal pathways which include; medial and lateral vestibulospinal tracts
- Termination; alpha and galma motor neurons and interneurons in laminae VII to IX within the spinal cord
- Function; their fibers control balance and posture by innervating the antigravity muscles- the flexors of the arm and extensors of the leg through the LMN

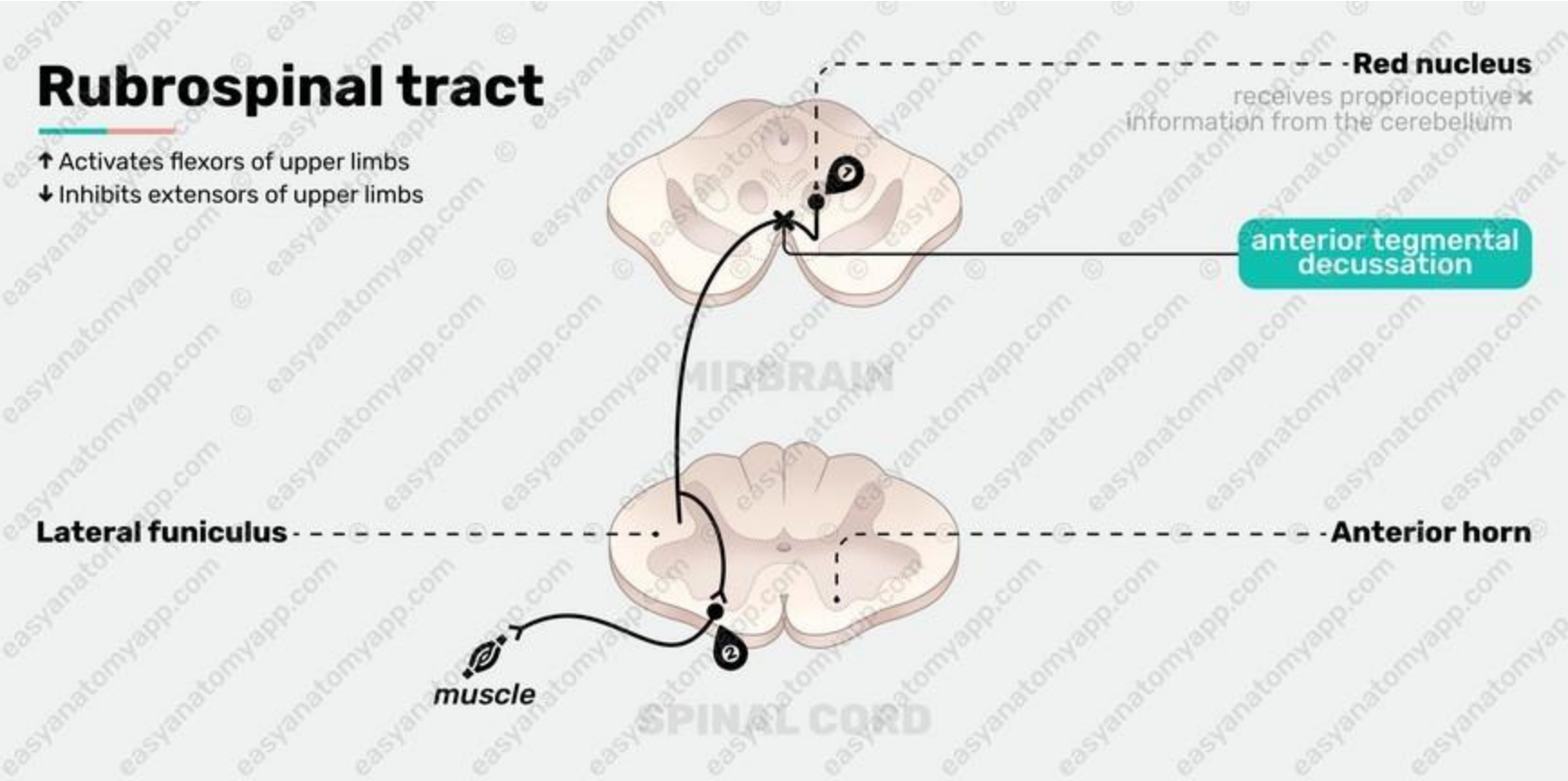
RETICULOSPINAL TRACTS

- Origin; arise from the reticular formations in pons and medulla
- This is made up of two tracts with different origins and functions
- Components;
- Corticoreticular fibers
- Pontine (medial) reticulospinal tract
- Medullary (lateral) reticulospinal tract
- Termination; anteromedial portion of laminae VII and VIII
- Functions; Muscle tone, balance maintenance, and postural changes, behavioral arousal, somatic motor control, cardiovascular control, pain modulation, sleep and consciousness, and habituation

RUBROSPINAL TRACTS

- This originates from the red nucleus in the midbrain tegmentum
- The fibers arise from the red nucleus and decussates to the opposite side of the CNS,
- They have contralateral innervations
- Termination; spinal cord at laminae V, VI and VII
- Function; regulation of flexors and extensor tone

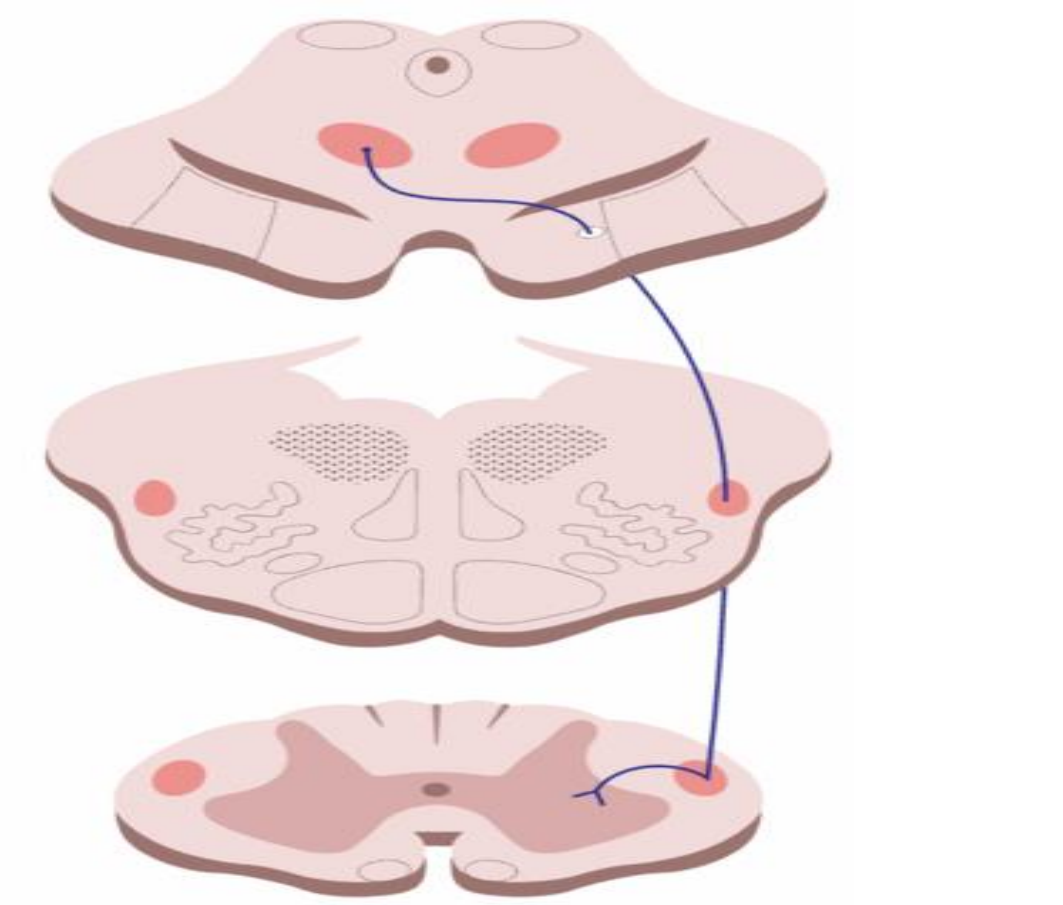
DIAGRAM


Rubrospinal tract
↑ Activates flexors of upper limbs
↓ Inhibits extensors of upper limbs
Red nucleus
receives proprioceptive information from the cerebellum
anterior tegmental decussation
MIDBRAIN
Lateral funiculus
Anterior horn
muscle
SPINAL CORD

TECTOSPINAL TRACTS

- Origin; fibers arise from the superior colliculus of the midbrain
- The superior colliculus receives input from the optic nerves, then decussates to the contralateral side and enters the spinal cord
- Termination; at the cervical levels of the spinal cord(neurons within laminae VI and VII)
- Function; it coordinates movements of the head in relation to visual and auditory stimuli

TECTOSPINAL TRACT



UPPER MOTOR NEURON(UMN)

- Located in the central nervous system (CNS)- the brain and spinal cord
- Originates in the motor areas and synapse with lower motor neurons
- Responsible for conveying signals that initiates and regulates voluntary movements

CAUSES OF UMN LESION

- Cerebrovascular accident (CVA)- stroke
- Traumatic brain injury
- Space occupying lesions- Malignancy
- Inflammatory disorders
- Neurodegenerative disorders
- infections

LOWER MOTOR NEURONES (LMN)

- Originates in the spinal cord and brainstem
- Directly innervates skeletal muscles
- Terminates in the peripheral nervous system (PNS)
- Responsible for executing signals received from the UMN
- Types- somatic motor neurons which cause skeletal muscle contraction and visceral motor neurons which controls smooth muscle, cardiac muscle, and glands
- Neurotransmitter- acetylcholine
- Lesions; affects fibers travelling from anterior grey horn to peripheral tissues,

CAUSES OF LMN LESIONS

- Amyotrophy lateral sclerosis (ALS)
- Poliomyelitis
- Spinal muscular atrophy (SMA)
- Stroke
- Neuropthies

CORTICOSPINAL TRACT

- Introduction ; the corticospinal tract is the largest and most important descending (pyramidal) tract of the motor pathway
- It is concerned with somatic movement (motor functions)
- Involved in skeletal muscle contractions
- Origin ; arises from the cerebral cortex in the-
    - i. primary motor area (Broadman area 4)
    - ii. Premotor area (Broadman area 6)
    - iii. Supplementary motor area or adjacent parietal area

The largest of the fibers are from pyramidal giant cells (of Bertz), some fibers also originates from the precentral gyrus and adjacent cortical areas

CORTICOSPINAL TRACT

- Course; the fibers descend downward and pass through the pyramid in the medulla
- Fibers from different parts of the cerebral cortex converge as they descend as cortico-fugal fibers which converges as corona radiate
- From the coronary radiata, the fibers descend through the posterior limb of the internal capsule
- The fibers concerned with cervical region is close to the genu (neck)
- The succeeding fibers to the upper limb is more posteriorly situated, and then in that descending order, those to the thorax, abdomen and lower limbs are more posterior(i.e they move progressively posterior)
- The fibers from the internal capsule goes to the crus cerebri intermediate 2/3rd and then passes through the pons further downward, then through the medulla where the fibers become more superficially situated, forming the pyramid

CORTICOSPINAL TRACT Course ont'd

- At the lower part of the medulla, slightly over 70% of the fibers decussate
- The decussated (crossed) fibers continues downward as the lateral corticospinal tract of the opposite spinal segment
- The uncrossed fibers constitute the anterior corticospinal tract
- A small fraction of the uncrossed fibers form the anterolateral corticospinal tract
- The pyramidal tract extend downwards to the most caudal part of the spinal cord
- As it descends, it decreases progressively in diameter
- More of the fibers leave to terminate in the spinal anterior grey horn

COURSE

- The anterior corticospinal tract occupies the adjacent portion between the anterior median fissure and anterior grey horn
- These fibers extend only up to the thoracic region (upper thoracic cord) and supplies primarily muscles of the upper extremity and the neck
- It is found only in humans and other higher animals like Apes
- The other uncrossed fibers called the anterolateral corticospinal tract constitute fibers that descend more anteriorly in the lateral funiculus
- About 55% of the pyramidal tract ends in the cervical part of the spinal cord
- About 20% ends in the thoracic region whereas the
- Remaining 25% ends in the lumbosacral segment

CLINICAL CONSIDERATIONS

- The pyramidal cells and their axons constitute what is known as the UMN while the anterior horn cells and fibers emanating from them to peripheral tissues form the LMN
- Damage to Corticospinal tract; the corticospinal tracts are susceptible to damage because they extend almost the entire length of the vertebral column
- Particularly vulnerable as they pass through the internal capsule which is a common site of CVA
- Unilateral lesion of left or right of corticospinal tract, will manifest symptoms on the contralateral side of the body

CLINICAL CONSIDERATION

- Damage to corticobulbar tract; Due to bilateral nature of the majority of the corticobulbar tracts, a unilateral lesion results in mild muscles weakness
- Spastic paralysis of the contralateral genioglossus (deviation of the tongue to the contralateral side) results from a lesion of UMN for hypoglossal nerve (CN-XII) while the LMN lesion will cause deviation to the damaged side
- Facial nerve; a lesion to the UMN for CN-VII will result in spastic paralysis of the muscles in the contralateral lower quadrant of the face
- Damage to the extrapyramidal tract; commonly seen in degenerative diseases, encephalitis and tumours
- They results in various types of dyskinesias or disorders of involuntary movement

UMN LESIONS

- EFFECTS;
- Hypertonia
- Hyper-reflexia
- Clonus
- Positive Babinski sign
- Increased muscle reflex
- Fasciculation absent
- Fibrillation absent

LMN LESIONS

- EFFECTS ;
- Flaccid paralysis,
- Loss of muscle tone and muscle wasting (atrophy)
- Babinski reflex is negative
- Hypotonia
- Fasciculation present
- Fibrillation present

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